

Kshitij Tripathi

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EDUCATION

VIT Bhopal University

Bachelor of Engineering in Computer Science

Aug. 2023 – Jul. 2027

CGPA: 8.53 / 10.0

EXPERIENCE

HaritBhoomi AgroCultivators LLP

Jun. 2024 – Dec. 2024

Application Development Intern · Bhopal, India

- Built a React + Node.js Farm ERP with IoT integration, reducing manual data-entry overhead by 65% for 3 farm supervisors across daily operations.
- Designed and deployed a REST API backend ingesting real-time IoT telemetry from 12+ sensor nodes, enabling live yield and inventory dashboards.
- Piloted at an ICAR-certified Farm of the Year; achieved a 40% improvement in worker efficiency and inventory utilization across a 6-week trial.
- Delivered iterative feature releases using Agile 2-week sprints, incorporating on-ground agronomist feedback into 4 production-ready versions.

Google Developer Groups on Campus (GDGC) — VIT Bhopal

Sep. 2025 – Present

Lead, Content & Design Team · Bhopal, India

- Directed a team of 8 to produce technical content and design assets, raising output consistency across 15+ published developer resources.
- Organized and co-managed 12+ workshops and campus tech events attended by 300+ students, covering Generative AI, cloud, and modern dev practices.
- Mentored 8 team members on technical writing, Figma, and content strategy, resulting in a 40% increase in event engagement quarter-over-quarter.

Cisco Community — VIT Bhopal

May 2024 – Present

Research Team Member · Bhopal, India

- Contributed to 6+ technical research documents on networking protocols, cybersecurity fundamentals, and emerging cloud technologies.
- Presented findings in 4 peer knowledge-sharing sessions, supporting continuous skill development for a community of 50+ students.

PROJECTS

FashionAI Assistant | FastAPI · TensorFlow · OpenCV · YOLOv8 · React

Dec. 2025

- Built an end-to-end AI fashion recommendation system; trained a TensorFlow CNN on 6,000+ labeled outfit images achieving 87% style-classification accuracy.
- Engineered a computer vision pipeline using OpenCV and YOLOv8 for real-time color harmony detection and garment segmentation across 12 style categories.
- Deployed a FastAPI REST API and a React + Tailwind CSS frontend serving occasion-specific outfit recommendations with sub-300 ms average response time.

F1 Driver Fingerprinting & Telemetry Analytics Platform.

Oct 2025

Python | Streamlit | FastF1 | Scikit-learn | UMAP | HDBSCAN | Machine Learning | Data Analytics

- Developed an end-to-end Formula 1 telemetry analytics platform to identify and classify unique driver behavior patterns using high-frequency racing telemetry data.
- Engineered machine learning pipelines for telemetry preprocessing, feature engineering, dimensionality reduction, and unsupervised clustering using UMAP and HDBSCAN.

- Generated driver “fingerprints” from braking intensity, throttle modulation, steering dynamics, cornering consistency, and gear-shift behavior for behavioral biometrics analysis.
- Implemented cross-track driver re-identification on unseen circuits, enabling robust driver recognition independent of track conditions.
- Built interactive Streamlit dashboards for telemetry visualization, comparative driver analysis, clustering insights, and performance exploration.
- Applied statistical analysis, feature extraction, and unsupervised learning techniques to support anomaly detection, behavioral analytics, and pattern recognition use cases.

Improved analytical interpretability and model scalability for real-time motorsport data exploration and transferable biometric identification applications.

SKILLINK — Student Collaboration Platform | JS · Random Forest · SBERT Oct. 2024

- Developed an ML recommendation engine using Random Forest regression and SBERT sentence embeddings; achieved 91% match-relevance score in user testing across 150+ student-researcher pairs.
- Designed and shipped the full frontend UI/UX as part of a 5-member Agile team, cutting mean time-to-match by 55% versus manual search.

Krishi Pradhan — Farm Management System | React · Node.js · MongoDB · REST API Nov. 2024

- Architected a full-stack farm management platform delivering real-time analytics on produce yield, inventory, and sustainable practice compliance for 30+ farm units.
- Integrated data-driven decision-making modules that improved yield forecasting accuracy by 23% and reduced input waste by 18% across tracked farm operations.
- Built 5 analytics dashboards visualizing 8+ crop and inventory KPIs, enabling farm managers to act on insights without manual reporting.

Stock Price Prediction Model | Python · Scikit-learn · Pandas Aug. 2024

- Built a predictive ML model for 15 NSE-listed equities using 5+ years of OHLCV data; applied feature engineering and hyperparameter tuning to achieve 79% directional accuracy.
- Reduced model inference latency by 40% through vectorized Pandas operations and caching of pre-computed feature windows.

TECHNICAL SKILLS

Languages: Python, C++, Java, JavaScript, Dart
Frameworks: React, Node.js, Express.js, FastAPI, Flutter, TensorFlow, Keras, OpenCV, YOLOv8
Databases: MongoDB, MySQL, Neo4j
Tools & DevOps: Git, GitHub, Docker, CI/CD, N8N Workflow Automation, Vite, Tailwind CSS
Cloud: AWS (EC2, S3, Lambda), Oracle OCI
Concepts: REST API Design, Agile / Scrum, Machine Learning, Computer Vision, Generative AI

CERTIFICATIONS

Oracle Cloud Infrastructure 2025 Certified Generative AI Professional	2025
Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate	2025
IBM — Introduction to Machine Learning	2024